

Image Processing Lab Projects

Project 1: Coin Detection & Classification

Goal: Develop a Python-based computer vision system that automatically identifies Euro coins from camera images and determines their total value.

Your task:

- Capture suitable images of a calibration object and Euro coins.
- Calibrate the camera to establish the relationship between image measurements and real-world dimensions.
- Develop your own approach to detect and measure coins in the images.
- Classify each coin based on its measured properties.
- Process multiple coins and calculate their total monetary value.
- Evaluate the accuracy and reliability of your approach.

Project 2: SMD Capacitor Detection

Goal: Develop a Python-based computer vision system that automatically detects and counts SMD capacitors on a graphics chip.

Your task:

- Capture a suitable image of the graphics chip.
- Develop your own approach to detect the capacitors, regardless of their orientation.
- Visualize the detected components and determine their total number.
- Evaluate and optimize your approach to achieve $\geq 90\%$ detection accuracy.

Project 3: Event Detection

Goal: Develop a Python-based computer vision system that automatically detects and visualizes events in a live camera stream.

Your task:

- Capture a suitable reference image under stable lighting conditions.
- Continuously analyze the live camera stream for relevant changes.
- Develop your own approach to distinguish actual events from irrelevant image variations.
- Automatically determine when an event has occurred.
- Isolate and visualize the detected event in the image.
- Evaluate and optimize your approach for reliable event detection.

Project 4: Gear Wheel Inspection

Goal: Develop a Python-based computer vision system that automatically inspects gear wheels for manufacturing defects and classifies specific error types.

Your task:

- Capture suitable images of reference gear wheels and defective samples under controlled lighting.
- Implement algorithms to detect and count individual gear teeth accurately.
- Identify and classify defects including missing, undersized, oversized, and fused teeth.
- Evaluate the accuracy and false-detection rate of your automated inspection approach.